

An End-to-End NLP Pipeline for Topic Detection in App Reviews

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Abstract

This work proposes an automated Natural Language Processing (NLP) pipeline for topic detection in mobile banking app reviews, using the HYPE S.p.A. app as a real-world case study. The proposed system aims to identify recurring issues, emerging trends, and customer pain points by leveraging state-of-the-art topic modeling techniques. The pipeline integrates advanced text preprocessing, embedding-based topic modeling using BERTopic, taxonomy alignment through semantic similarity and multi-label topic assignment. Experimental results demonstrate that the proposed approach enables scalable, consistent, and semantically meaningful analysis of large volumes of app reviews, supporting data-driven decision-making in product development and customer experience management. The source code is available at <https://github.com/adsp-polito/2025-P1-Topic-detection/>

1 Introduction

User-generated app reviews constitute a valuable source of information for identifying usability issues, and prioritizing product improvements. In the fintech domain, this feedback is particularly critical, as defects affecting authentication, transactions, or customer support can directly impact user trust. However, the large volume, unstructured nature, and linguistic variability of app reviews make manual analysis impractical at scale.

To address this challenge, automated text analysis techniques are increasingly adopted to extract actionable insights from large collections of user feedback. Nevertheless, applying such techniques to real-world app reviews remains challenging due to multi-topic semantics, multilingual data, and the strong impact of preprocessing choices on result quality. Moreover, for enterprise adoption, the extracted insights must be interpretable and aligned with existing domain taxonomies to enable integration into established monitoring workflows.

A flexible topic detection pipeline, developed in collaboration with the Italian fintech HYPE S.p.A., is introduced to handle these complexities. By leveraging BERTopic, this comprehensive pipeline enables the systematic analysis of industrial app reviews. The main contributions of this work are as follows: (1) a robust preprocessing pipeline for app review analysis, including multilingual handling and domain-specific stopword strategies, (2) an extensive experimental analysis of BERTopic configurations, comparing different architectural variants and execution modes, (3) a multi-label topic assignment method based on topic probability distributions, and (4) a semantic taxonomy alignment framework for mapping data-driven topics to an existing domain taxonomy.

To validate the effectiveness of the proposed pipeline, this work addresses the following research questions:

- **RQ1:** Does machine translation affect the semantic quality of topic detection on multilingual app reviews?

- **RQ2:** How do different stopword-removal strategies affect the performance of embedding-based topic modeling?
- **RQ3:** Which configuration of BERTopic’s modular components yields the best topic quality?
- **RQ4:** Do semi-supervised topic models outperform unsupervised approaches in identifying relevant issues?
- **RQ5:** Do embedding-based semantic similarity methods provide reliable mappings between data-driven topics and the existing taxonomy of app issues?

The remainder of the paper details the proposed methodology and its experimental evaluation.

2 Related Work

User feedback in app store reviews has been widely analyzed as a rich source of user opinions and issues [11], and NLP techniques, such as topic modeling, have emerged as powerful tools to analyze reviews and extract user feedback [18]. Conventional topic modeling approaches, including Latent Dirichlet Allocation (LDA) [7] and Non-Negative Matrix Factorization (NMF) [17], have been widely adopted in the past years. However, prior studies [14, 15] have highlighted that standard LDA performs poorly on short-text corpora, primarily due the lack of word co-occurrence information.

To address these limitations, recent advances have introduced embedding-based approaches like BERTopic [12] which are better suited to capture the semantic context of short texts. Comparative studies show that these models outperform traditional topic modeling approaches [9].

Both supervised and unsupervised machine learning approaches have been proposed to extract informative feedback from user reviews [4, 10, 16]. However, this work differs by analyzing the performance of various configurations of a single topic modeling framework to support different operational requirements.

Although preprocessing techniques are widely used in sentiment analysis, their application to user review analysis has been less explored. The effectiveness of stopword removal remains debated. While some methods incorporate it [12], recent research [19] suggests it may degrade performance, highlighting the need for further evaluation in app review analysis as well.

3 Methods

3.1 Dataset and Preprocessing

3.1.1 Dataset and Reference Taxonomy Description. The dataset consists of 6,150 real app reviews collected from multiple channels and provided by the industrial partner. Each record is associated with a unique identifier and includes the review text, a user rating, a sentiment label assigned by the company’s existing classification system, and, when available, a reference label obtained either automatically or through expert manual annotation, according to their

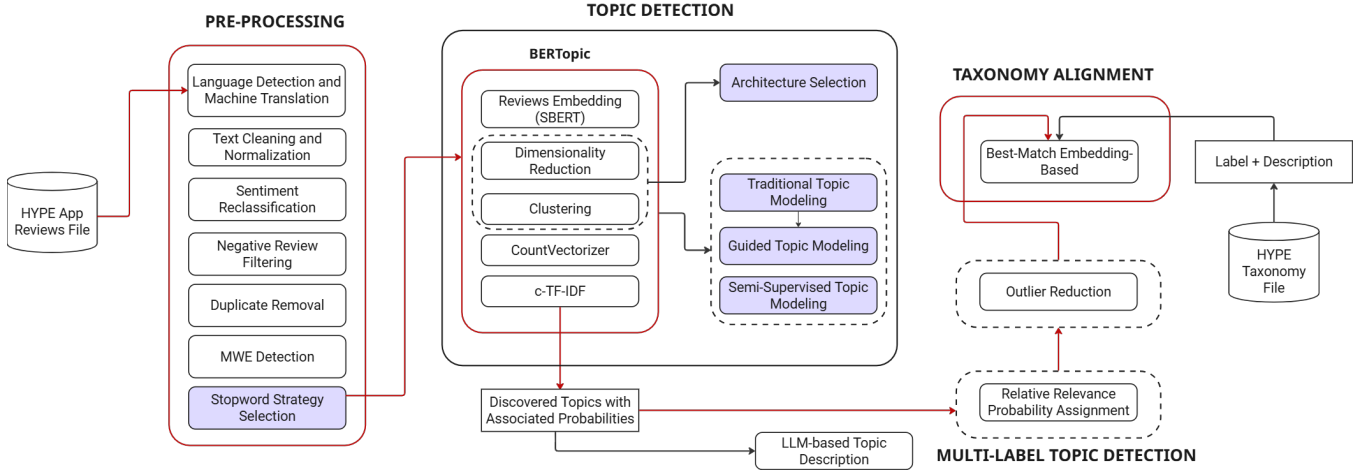


Figure 1: Overview of the proposed topic detection pipeline. Red lines represent the core steps, while black and dotted blocks show optional components explored as part of the research. Components highlighted in purple denote configurable stages.

taxonomy. Additional metadata, such as model version and review timestamp, are also included (schema in Appendix A).

Furthermore, this study relies on an issue taxonomy provided by HYPE, which defines a structured set of problem categories used to label user complaints within the application domain. Each taxonomy entry corresponds to a semantic issue class, characterized by a unique label and associated textual descriptors. The taxonomy is organized hierarchically, allowing fine-grained issue categories to be optionally grouped under higher-level parent classes. This taxonomy serves as a reference framework for aligning automatically discovered topics with established issue categories (Appendix A).

Together, these datasets provide the foundation for the topic modeling and evaluation procedures described in the following sections, acting as the input layer of the pipeline shown in Figure 1.

3.1.2 Language Detection and Machine Translation (RQ1). The review dataset is inherently multilingual, which may hinder topic coherence and the application of language-specific techniques. To address this, automatic language detection is applied to identify non-Italian reviews, using a transformer-based model built on XLM-RoBERTa [8] and fine-tuned for multilingual language identification [21]. This step identifies 206 non-Italian reviews out of 6,150, with English and Portuguese being the most prevalent languages.

All non-Italian reviews are subsequently translated into Italian using a multilingual neural sequence-to-sequence model based on the No Language Left Behind (NLLB) [20] architecture. The translated texts are then merged with the original Italian reviews and jointly processed by the pipeline.

3.1.3 Text Cleaning and Normalization. Text preprocessing follows an ordered pipeline aimed at reducing noise and standardizing the textual content prior to topic modeling. Standard normalization steps are applied, including whitespace normalization, character repetition reduction, URL and user mention removal, emoji-to-text conversion, and lowercasing applied to the sentiment column.

Neutral reviews are reclassified using a transformer-based sentiment ensemble to mitigate uncertainty, motivated by the empirical

observation that several reviews originally labeled as neutral still contain explicit user complaints. The ensemble combines three complementary components: FEEL-IT [6] used in two inference configurations and XLM-RoBERTa fine-tuned for multilingual sentiment modeling [5]. Model outputs are mapped to a numeric sentiment scale and aggregated into a composite score, from which the final label is derived through a decision rule that intentionally favors negative sentiment. Only reviews classified as negative are retained for subsequent steps.

After sentiment-based filtering, meaningless (e.g., “ok”, “bene”, “..”) and redundant reviews are removed. Duplicate detection is performed using a cosine similarity criterion based on TF-IDF to prevent bias introduced by repeated content.

Finally, multi-word expressions (MWEs) are identified to preserve domain-specific concepts. Candidate MWEs are extracted using a PMI-based statistical collocation approach with part-of-speech constraints, retaining frequent noun-noun and noun-adjective bigrams, which are merged into single tokens to improve topic interpretability.

Overall, these steps ensure a clean and consistent textual representation for topic detection.

3.1.4 Stopword Strategies (RQ2). Stopword removal can significantly influence topic modeling performance, particularly in domain-specific corpora where highly frequent terms may have limited discriminative power. Accordingly, this study evaluates the impact of different stopwords strategies, including a domain-specific list automatically derived using a TF-IDF-based approach inspired by previous work [23].

The domain-specific stopwords list is constructed from a large publicly available corpus of banking app reviews [1], translated into Italian for linguistic consistency and processed to extract terms with consistently high TF-IDF scores across documents. The top 200 ranked terms are retained to form a stopwords list tailored to the banking application domain.

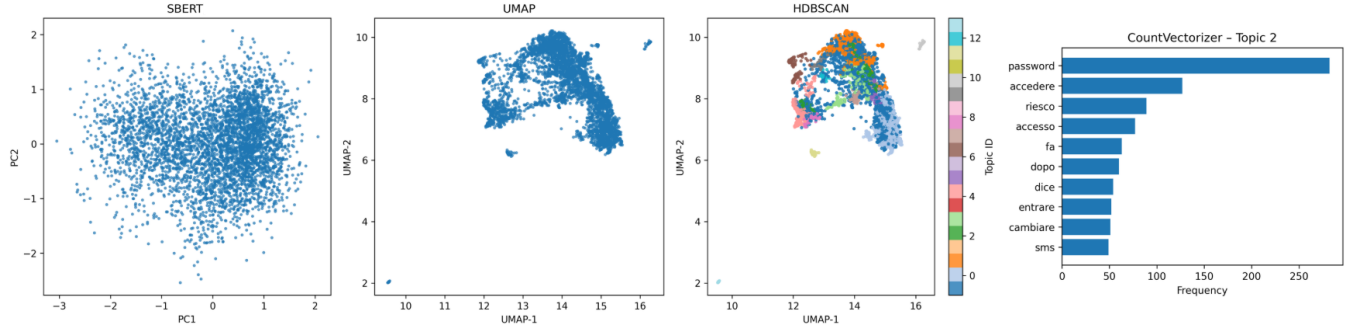


Figure 2: BERTopic workflow, related to the topic characterized by the keywords ['password', 'accedere', 'cambiare', 'riesco', 'accesso', 'entrare', 'errore', 'dice', 'fa', 'posso']"

Based on this setup, the following stopwords strategies are evaluated as part of Research Question RQ2:

- **No Stopword Removal**, where all terms are retained;
- **Generic Stopword List**, using a standard language-dependent stopwords set;
- **Domain-Specific Stopword Removal**, applying only the automatically extracted domain-specific stopwords;
- **Delta Stopwords**, removing terms present in the domain-specific list but not in the generic one;
- **Union Stopwords**, combining both generic and domain-specific stopwords.

3.2 System Overview and Processing Pipeline

The proposed system applies an end-to-end topic detection pipeline to automatically identify the main issues reported by users and aligning the inferred topics with the reference taxonomy provided by the industrial partner.

The pipeline consists of three stages: (i) topic detection through an embedding-based topic modeling framework, where multiple models and configurations are evaluated to identify the most effective setup; (ii) multi-label topic assignment to capture reviews addressing multiple issues simultaneously; and (iii) taxonomy alignment using embedding-based similarity, aimed at qualitatively assessing the discovered topics and steering them toward the HYPE taxonomy.

3.3 Topic Detection Framework

The topic detection framework evaluates both classical and embedding-based approaches. Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF) are considered as baseline methods to establish a benchmark, while BERTopic is adopted as a more advanced embedding-based topic modeling approach.

3.3.1 Baseline methods: LDA and NMF. LDA [7] is a generative probabilistic model of a text corpus that represents documents as random mixtures of latent topics, where each topic is characterized by a distribution over words. NMF [17] is a matrix factorization algorithm that finds the positive factorization of a given non-negative matrix by decomposing it into two lower-rank non-negative matrices, representing topic-term and document-topic relationships.

3.3.2 BERTopic. BERTopic [12] is a topic modeling framework that leverages document embeddings, clustering techniques and a class-based variation of TF-IDF to generate coherent topic representations. The topic representations are obtained through a sequence of fixed steps, as shown in Figure 2.

- (1) **Document embeddings.** Each review is encoded using a pre-trained transformer-based model; in this work, a Sentence-BERT (SBERT) [2, 22] model is employed.
- (2) **Document clustering.** Due to the high dimensionality of document embeddings, dimensionality reduction is first applied using UMAP, followed by clustering with HDBSCAN, a density-based algorithm with a hierarchical formulation.
- (3) **Topic Representation.** Topic representations are built at the cluster level using a bag-of-words approach. For each cluster, a CountVectorizer is applied to the reviews assigned to that cluster, incorporating Italian stopwords removal based on the NLTK stopwords list and extended with project-specific noise terms (e.g., "app", "hype", "banca", "stelle") to avoid their dominance in the extracted topic keywords.

The resulting term frequencies are used to compute the class-based TF-IDF (c-TF-IDF), which highlights terms that are representative of individual topics by treating all documents within a cluster as a single aggregated document:

$$W_{t,c} = \text{tf}_{t,c} \cdot \log \left(1 + \frac{A}{\text{tf}_t} \right) \quad (1)$$

where $\text{tf}_{t,c}$ denotes the frequency of term t in cluster c , tf_t its frequency across all clusters, and A the average number of words per cluster. This weighting scheme emphasizes terms that provide high discriminative power for a given topic.

Finally, a Large Language Model (LLM), specifically LLaMA-3.1 (8B Instruct) [3], is used to generate business-operational descriptions from the topics discovered (results shown in Appendix C).

3.4 Topic Configuration and Modeling Variants

To assess the impact of different design choices on topic representation quality, this work evaluates multiple topic modeling configurations along two dimensions: model architecture and degree of supervision.

3.4.1 Architecture selection (RQ3). BERTopic provides a modular architecture that allows different algorithms to be combined across its main components. In this work, multiple model configurations are evaluated by varying both the dimensionality reduction and clustering techniques.

For dimensionality reduction, UMAP and Kernel PCA are employed: UMAP focuses on preserving local neighborhood structure in high-dimensional embeddings, while Kernel PCA captures non-linear patterns while maintaining global data structure.

For clustering, K-means, spectral clustering, and HDBSCAN are evaluated: K-means performs centroid-based partitioning with a fixed number of clusters, spectral clustering exploits graph-based similarities to capture non-convex structures, and HDBSCAN applies density-based clustering with automatic cluster selection and outlier handling.

3.4.2 Semi-Supervised and Guided Topic Modeling Approaches (RQ4). In addition to the traditional topic modeling pipeline described in 3.3.2, this work investigates guided and semi-supervised variants of BERTopic. In the traditional setting, topic discovery is entirely unsupervised, while document embeddings are obtained from pre-trained language models.

Guided topic modeling is implemented by providing BERTopic with a predefined set of 15 domain-specific seed topics, each defined as a small list of representative keywords inspired by the application domain (e.g., related to authentication, payments, support, or bugs) and manually identified to achieve a balance level of topic granularity and interpretability within the provided taxonomy (see Appendix B for the complete list). These seed topics are passed to the model through the *seed_topic_list* parameter. The keywords of each seed topic are concatenated and embedded in the same semantic space as the reviews. Each review is then associated with the most similar seed topic based on cosine similarity, while reviews with no sufficient similarity remain unlabeled. These soft labels are used to guide topic formation during dimensionality reduction, and seeded keywords are further emphasized during topic representation by increasing their weights in the c-TF-IDF computation, making them more likely to appear among the topic keywords.

Semi-supervised topic modeling guides the embedding space toward a predefined set of labels by encouraging reviews sharing the same label to be grouped within the same cluster. Supervision is derived from the HYPE reference dataset by retaining only reviews associated with a single label and labels with at least 25 occurrences. From this filtered subset, 15% of the reviews are randomly selected as supervised examples, a proportion chosen based on the dataset size, the BERTopic documentation, and best practices in weak supervision. The remaining reviews are treated as unlabeled and assigned a value of -1 . These labels are encoded into a supervision vector y and provided as input to UMAP to softly steer topic formation toward the predefined categories.

3.5 Multi-Label Topic Detection

To overcome the limitations of single-topic assignment, this study adopts a multi-label strategy based on BERTopic topic probability distributions. For each document d , let

$$p_{\max} = \max_k p_{d,k} \quad (2)$$

denote the highest topic score in the corresponding topic probability distribution. A topic k is selected as a candidate label if

$$p_{d,k} \geq \alpha \cdot p_{\max}, \quad (3)$$

where $\alpha \in (0, 1)$. As a result, only topics whose contribution is comparable to the most representative one are retained. To limit topic fragmentation, the number of assigned labels is further constrained by a maximum label cardinality $L_{\max}$. In our experiments, the relative confidence threshold was set to $\alpha = 0.85$, and the maximum number of assigned labels was limited to $L_{\max} = 3$.

The same distribution-based criterion was also employed to reduce the number of outliers by reassigning documents to the second most probable topic whenever its membership probability satisfied $p_{d,k(2)} \geq 0.20$. These values were selected empirically to balance topic coverage and interpretability.

3.6 Taxonomy Alignment Strategy

To assess the semantic correspondence between the topics discovered by BERTopic and the predefined taxonomy, taxonomy alignment is performed using semantic similarity in a shared embedding space. Each discovered topic t is represented using the standard textual representation provided by BERTopic and encoded into a vector $\mathbf{e}_t \in \mathbb{R}^d$. Similarly, each taxonomy label ℓ is represented through a predefined textual description combining the label name and its associated description, and embedded into a vector $\mathbf{e}_\ell \in \mathbb{R}^d$ using the same embedding model.

For each topic t , similarity scores with all taxonomy labels are computed as

$$s(t, \ell) = \text{cosine}(\mathbf{e}_t, \mathbf{e}_\ell), \quad (4)$$

The taxonomy label assigned to topic t is obtained by selecting the label that maximizes the similarity score:

$$\ell^*(t) = \arg \max_{\ell \in \mathcal{L}} s(t, \ell), \quad (5)$$

where $\mathcal{L}$ denotes the set of taxonomy labels.

This alignment procedure provides an interpretability and evaluation layer for assessing the correspondence between discovered topics and the reference taxonomy, while its reliability (RQ5) is analyzed in the experimental section.

4 Experiments

This section details the experimental validation of the proposed pipeline. The analysis focuses not only on quantitative topic modeling metrics, but also on the system's ability to address the underlying business objective of identifying actionable issues in mobile app reviews.

4.1 Evaluation Metrics

To quantitatively assess topic quality, three complementary metrics are employed.

4.1.1 Silhouette Score. This measures the density and separation of the clusters. For a data point i , let $a(i)$ denote the average intra-cluster distance, and $b(i)$ the average distance to the nearest neighboring cluster. The silhouette score is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (6)$$

The mean score ranges from $[-1, 1]$, with higher values indicating better-separated and more compact clusters.

4.1.2 Topic Coherence Score. Topic coherence measures the semantic consistency of a topic based on the similarity of its top representative words. An embedding-based coherence metric is adopted. For a given topic t , let P denote the set of all pairwise combinations of its top representative words (w_i, w_j) , and let $\mathbf{e}_w$ denote the embedding vector of word w :

$$\text{Coherence}(t) = \frac{1}{|P|} \sum_{(w_i, w_j) \in P} \text{cosine}(\mathbf{e}_{w_i}, \mathbf{e}_{w_j}) \quad (7)$$

The coherence score ranges from $[-1, 1]$, with higher values indicating more semantically coherent topics.

4.1.3 Topic Diversity Score. This metric assesses redundancy across topics by measuring the proportion of unique words among all topic descriptors. Let $|W|$ be the number of unique words across topics, T the total number of topics, and K the number of top words per topic. The score is defined as:

$$\text{TopicDiversity} = \frac{|W|}{T \cdot K} \quad (8)$$

The diversity score ranges from $[0, 1]$, where higher values indicate lower lexical overlap across topics.

4.2 Experimental Setup

All experiments were conducted on the subset of negative app reviews, using BERTopic as the primary topic modeling framework, with LDA and NMF as baseline models. Two BERTopic configurations were tested: the default configuration for baseline comparisons, and a best-performing one selected based on evaluation metrics and outlier proportion, as shown in Table 1.

From a business perspective, a limited number of topics is preferred for better interpretability and adoption. While the solution allows flexible topic control via the *reduce_topics* procedure, experimental results are based on a fixed number of 15 topics to ensure fair comparison across models.

LDA and NMF were configured with the same number of topics and default hyperparameters. The outlier reduction strategy described in Section 3.5 was applied to all experiments involving BERTopic (UMAP + HDBSCAN), while was not used in the baseline models for compatibility reasons.

Table 1: Dimensionality reduction and clustering hyperparameters evaluated in the experimental setup.

Component	Parameter	Default	Best
<i>Dimensionality Reduction</i>			
UMAP	n_neighbors	15	20
UMAP	n_components	2	5
UMAP	min_dist	0.1	0.0
UMAP	metric	euclidean	cosine
<i>Clustering</i>			
HDBSCAN	min_cluster_size	5	20
HDBSCAN	min_samples	None	5
K-means	n_clusters	15	–
Spectral Clustering	n_clusters	15	–

4.3 Results

This section presents the experimental results, starting with a comparison of BERTopic to baseline models to justify its selection. It then details experiments addressing the defined research questions.

4.3.1 Baseline Comparison. The study compares the proposed BERTopic approach with two baseline topic modeling methods: LDA and NMF.

As shown in Table 2, although LDA achieves the highest coherence score, it exhibits very low topic diversity, indicating lexical overlap across topics. Qualitative inspection confirms that LDA topics frequently reuse generic terms (e.g., *app*, *problem*, *update*), resulting in weakly discriminative clusters.

In contrast, BERTopic achieves the highest topic diversity, producing more fine-grained and distinct topics. While its coherence score is lower, the resulting topics better separate specific technical issues, such as authentication failures and card delivery problems.

Table 2: Comparison of topic modeling approaches.

Model	Coherence	Diversity
LDA (Baseline)	0.6632	0.3333
NMF (Baseline)	0.6190	0.6867
BERTopic (Ours)	0.5393	0.8071

4.3.2 Impact of Machine Translation (RQ1). To assess the impact of machine translation in the context of topic modeling, cluster quality is evaluated on the subset of translated reviews. In particular, the silhouette score was computed by calculating the silhouette value $s(i)$ for each translated review, and then averaging them. Results show that the silhouette score obtained on translated reviews (0.068) is comparable to the score computed on the full dataset (0.0636), indicating that machine translation does not degrade cluster compactness or separation.

In addition, the proportion of correctly predicted labels for translated reviews is measured using the mono-setting alignment-based precision metric defined in Section 4.3.6. Precision increases from 0.1950, measured on all reviews, to 0.2031 when evaluated on reviews that were originally written in foreign languages.

4.3.3 Stopword Strategy Analysis (RQ2). The different stopwords strategies presented in Section 3.1.4 were assessed for their impact on topic modeling performance. As shown in Table 3, the *No Stopword Removal* strategy yielded the highest values for both silhouette and coherence, indicating that retaining all terms resulted in the most coherent and well-separated topics.

Table 3: Comparison of stopwords strategies. Results computed using the default approach and architecture.

Strategy	Silhouette	Coherence	Diversity
No Stopword Removal	0.0636	0.5375	0.8071
Generic Stopword List	0.0508	0.5313	0.8000
Domain-Specific Stop.	0.0427	0.4993	0.8786
Delta Stopword	0.0389	0.5044	0.8857
Union Stopword	0.0434	0.5236	0.8714

Although the highest diversity was achieved with the *Delta Stopword* strategy, the *No Stopword Removal* strategy still maintained a strong diversity. These results can be attributed to BERTopic’s use of contextual embeddings, in which retaining stopwords helps preserve key contextual information, thereby enhancing topic quality.

4.3.4 BERTopic architecture comparison (RQ3). To determine the best BERTopic configuration, different combinations of dimensionality reduction and clustering methods were tested. Table 4 shows the results of the tested combinations, while Table 1 details the default hyperparameters. The original BERTopic architecture employing UMAP and HDBSCAN provided the best balance between topic separation, diversity, and coherence in the context of the evaluated combinations. This combination likely works best because UMAP preserves data structure effectively, while HDBSCAN captures well-separated, diverse topics through density-based clustering.

Table 4: Comparison of dimensionality reduction and clustering combinations. Results computed using the default approach and hyperparameters, without stopwords removal.

Architecture	Silhouette	Coherence	Diversity
UMAP + HDBSCAN	0.0385	0.5393	0.8071
Kernel PCA + Spectral	-0.016	0.6919	0.5200
Kernel PCA + K-Means	0.0345	0.5691	0.5267
UMAP + Spectral	0.0198	0.5470	0.5733
UMAP + K-Means	0.0147	0.5681	0.5400

4.3.5 Effect of Supervision on Topic Modeling Performance (RQ4). To investigate whether supervision in topic modeling leads to better results in identifying relevant issues compared to traditional approaches, the three different methods explained in Section 3.4.2 were compared. Results in Table 5 show that the Traditional topic modeling approach performed slightly better than the semi-supervised and guided approaches across all metrics, although the differences among the three methods were relatively small. However, supervision in topic modeling can still be a valuable asset for HYPE in the future, particularly when analyzing new reviews. In such cases, supervision can help steer the topic discovery process towards pre-defined labels, enhancing the relevance and focus of the results while ensuring consistency with previous analyses.

Table 5: Comparison of BERTopic execution modes. Results are computed using the default architecture, without stopwords removal.

Execution Mode	Silhouette	Coherence	Diversity
Traditional	0.0636	0.5375	0.8071
Semi-supervised	0.0559	0.5336	0.8071
Guided	0.0499	0.4969	0.8071

4.3.6 Taxonomy Alignment Assessment with Precision (RQ5). To evaluate taxonomy alignment, example-based precision is computed under two settings: *mono-label* and *multi-label*, enabling the assessment of the proposed multi-label classification strategy. Let $\hat{Y}_i^{(m)}$ denote the set of predicted taxonomy labels for review i under evaluation mode $m \in \{\text{mono}, \text{multi}\}$, and let Y_i denote the corresponding ground-truth label set. Example-based precision [13] is defined as:

$$\text{Precision}^{(m)} = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{Y}_i^{(m)} \cap Y_i|}{|\hat{Y}_i^{(m)}|}$$

where N is the total number of reviews. Reviews with no predicted or ground-truth labels are assigned to a dedicated *outlier* class to ensure numerical stability.

Using this metric, the proposed approach achieves a precision of 0.1950 in the mono-label setting and 0.1820 in the multi-label setting, indicating that allowing multiple label assignments preserves a comparable level of correctness.

Using a semantic-based approach is considered valid despite the results, as the ground truth labels used are obtained from other automatic or manual systems. Consequently, a qualitative check highlights that correct labels may occasionally be marked as incorrect due to inconsistencies in the ground truth.

5 Conclusion

This work introduced an automated NLP pipeline for topic detection in mobile banking app reviews, with the objective of extracting coherent and actionable topics to support customer feedback analysis. Through the evaluation of multiple configurations, the optimal setup was identified, combining UMAP for dimensionality reduction and HDBSCAN for clustering, without stopwords removal and using default execution mode. This configuration produced 14 coherent topics with an outlier rate of 19.12%, effectively balancing noise reduction and the preservation of meaningful, actionable feedback for practical applications.

The moderate values obtained by standard topic modeling metrics can be attributed to the small dataset size and to the intrinsic characteristics of app review data, which are typically short, noisy, and semantically heterogeneous. Moreover, reviews often exhibit an inherent multi-topic nature, with individual entries simultaneously addressing multiple issues, which makes topic detection through clustering particularly challenging. This difficulty is further amplified when aligning discovered topics with HYPE’s fine-grained taxonomy. Such properties limit cluster compactness and separability even when contextual embeddings are employed. Furthermore, evaluation metrics such as coherence and silhouette primarily capture geometric or lexical properties and are not fully aligned with the business objective of identifying actionable issues.

There are several avenues for improvement and future research. These include exploring hierarchical topic modeling to gain more granular insights from large and complex datasets, further optimizing performance through hyperparameter tuning, and experimenting with Italian-specific sentence embeddings to improve topic quality.

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6 Appendices

This appendix contains supplementary information related to the experiments described in the main body of the document.

A Dataset Schema

This section provides an overview of the structure and fields of the datasets used for the experiments. The following tables describe the key attributes and data types for each dataset.

Table 6: Schema of the app review dataset used in the experiments.

Field name	Type	Description
id_reviews	Integer	Unique identifier
review	Text	Raw review text
rating	Integer (1–5)	User satisfaction rating
sentiment	Categorical	Positive / Neutral / Negative
labels	List	Topic category assigned
is_labels	Boolean	Label availability
model_version	Categorical	Processing model version
review_date	Timestamp	Review publication date

Table 7: Schema of the taxonomy dataset.

Field Name	Type	Description
label	Text	Topic or label name
parent_label	Text	The parent label associated with the topic
description	Text	Detailed description of the label
old_label_name	Text	Previous label name if updated
deprecated	Boolean	Whether the label is deprecated or not

B Seed Topics for Topic Modeling

This section outlines the seed topics used in the topic modeling experiments. These seed topics represent key areas of interest based on domain-specific terms related to various features and services within the app. The following topics were defined, according to the taxonomy, to guide the identification of relevant themes in the app reviews:

- **Access and Authentication:** "accesso", "login", "password", "codice verifica", "biometrico", "credenziali", "dispositivo"
- **Transfers and Payments:** "bonifici", "bonifico", "istantanei", "periodici", "trasferimento"
- **Card and Shipping:** "carta", "spedizione", "design", "rinnovo", "materiale", "blocco"
- **Costs and Fees:** "costi", "commissioni", "bancario", "elevate"
- **Online Payments:** "pagamenti", "online", "siti web", "esecuzione", "ricariche"
- **Withdrawals:** "prelievi", "prelevare", "restrittivi", "atm"
- **Account and Deposits:** "ricarica", "conto", "versamento"
- **Plans and Subscriptions:** "piano", "downgrade", "upgrade", "gratuito", "premium", "business"
- **Errors and Bugs:** "bug", "errori", "rallentamenti", "malfunzionamenti", "lenta"
- **Support and Assistance:** "assistenza", "supporto", "segnalazioni", "operatori", "telefonica"
- **Fraud Prevention:** "blocco", "conto", "antifrode", "immotivato"
- **Rewards and Bonuses:** "cashback", "bonus", "premi", "concorsi"
- **Notifications:** "notifiche", "push", "email", "comunicazione"
- **Digital Wallets:** "wallet", "apple pay", "google pay", "virtuale"
- **Investments and Cryptocurrencies:** "investimenti", "grafici", "bitcoin", "crypto"

C Topic Modeling Results

This section presents the results of the topic modeling experiments, including the generated topics, their descriptions, top words, the best-matching taxonomy labels, and the similarity scores computed between the top words of each discovered topic and the corresponding taxonomy labels, considering also its textual descriptions. The table below summarizes the key information:

Table 8: Topic Modeling Results

ID	Topic Description	Top Words	Best Match Label	Similarity Score
0	Problemi con l'app: cartelle visualizzate deformate e problemi d'accesso ripetuti.	accedere aggiornamento riesco funziona fa apre carta dopo entrare giorni	problemi_accesso	0.648400009
1	Problemi con l'invio della nuova carta e la sostituzione della vecchia.	carta soldi assistenza dopo stato ancora senza pagamento bloccato giorni	problemi_gestione_carta	0.603699982
2	Problemi con l'accesso all'app e al sito web. Utenti lamentano difficoltà nell'accedere ai loro conti.	password accedere cambiare riesco accesso entrare errore dice fa posso	problemi_accesso	0.766900003
3	Problemi con l'aggiornamento dei dati personali.	video registrazione aggiornamento fa dati documento accedere riesco dopo dice	problemi_dati_conto	0.656199992
4	Problemi con il sistema che causa blocchi e mancanza di contatto con l'assistenza.	operatore chat parlare bot servizio_clienti assistenza mail via attesa problema	problemi_assistenza	0.734899998
5	Problemi con i pagamenti e le transazioni bancarie.	fa bonifici funziona mai problemi revolut ogni bonifico blocca va	problemi_pagamenti_generici	0.694599986
6	Problemi con l'accesso biometrico che non riesce ad autenticarsi correttamente.	accesso_biometrico face id valido riconoscimento funziona biometrico accedere riconosce password	problemi_accesso	0.612100005
7	Problemi con l'esperienza d'uso: app sembra essere lenta ed inefficiente.	lenta molto lentezza troppo caricamenti caricamento macchinosa diventata apertura lentissima	problemi_download_app	0.617399991
8	Problemi relativi alle notifiche push che includono pubblicità e non sono pertinenti.	notifiche pubblicità spam promozioni disattivare notifiche_pubblicitarie notifiche_push arrivano troppa possono	problemi_comunicazione	0.550000012
9	Problemi con l'accesso e assistenza clienti sono considerati gravi.	assistenza servizio pessima assistenza_inesistente scadente assente scarsa peggiore inesistente essenziale	problemi_assistenza	0.677900016
10	Problemi con la carta che non mantiene lo stato magnetico e presenta difficoltà nella visualizzazione dei soldi.	carta ricaricare nuova altra problemi design ricarica funziona verticale modo	problemi_gestione_carta	0.457100004
11	Problemi relativi alla gestione e trasferimento dei fondi (bitcoin) nel portafoglio.	bitcoin vendere portafoglio connessione euro conio problema inviare invio	problemi_circuito_pagamento	0.681200027
12	Problemi con l'accesso al proprio account.	questionario dato do tolgo problema mettere errore settimana ottimo plafond	problemi_accesso	0.665700018
13	Problemi con il servizio clienti che non risponde alle richieste degli utenti.	numero opzione cambiato logicamente senza chiamare evitare arrabbiato schifo professionali	problemi_servizi_partner	0.679199994